Chapter 6

Learning Advanced Prompting

IN THIS CHAPTER

- **Prompting with the result in mind**
- **Understanding all the ways to target and add data**
- **Adjusting the AI's approach to interpreting data**
- **Changing data influence to direct responses**

In Chapter 2 of this mini-book you learned basic and intermediate prompting skills. In this chapter, you learn advanced prompting skills and some unexpected ways to boost the power in your prompts by using memory features and temperature settings. Don't worry. It's not as hard as it may sound. You can start using these techniques right away, too!

Starting at the End: Defining Desired Outputs before Prompting

If you want to make sure that GenAI results take the direction you want, give it a clear sense of the destination. After all, it's impossible to plan the best path to take if you have no idea where you want to go.

Even if you're using GenAI to explore a topic or dive deep into data discovery, you should be clear about your reason for doing so. In other words, have a point to the exercise before prompting. Otherwise, you'll find yourself adrift, riding the wind or an intellectual whim, and in dan-

ger of spending hours falling down the notorious virtual rabbit holes. Of course, rabbit holes are okay if you have time to wander about.

But for most, GenAI is a tool to get work done. If that's what you use it for, you'll want to take the most efficient and direct path to your end goal. For example, if you're looking for a comprehensive explanation of a scientific concept, it's important to specify not just the concept itself but the depth of the explanation you need, the context in which you're going to use the information, and any aspects you're interested in. This level of detail in the prompt guides a chatbot to generate a more fitted and useful response. Keep in mind that adding this level of detail to one or more prompts requires some thought and planning. Rarely will you or anyone else be able to capture all these details on a whim. It's easy to forget or overlook an important element in the moment.

To fully grasp this concept of working backward from result to query — or in the case of GenAI from response to prompt — consider why you're doing this exercise at all. Odds are that you've already formulated certain elements like these:

- A specific idea
- A concept
- An execution method
- Formats or mediums
- The target audience
- A specific message

Combining those elements (however many or few) defines your end goal, which in turn informs and defines what you ultimately want in an AI response. Start there.



REMEMBER When you have an end vision in mind, start planning your prompts in a way that points you toward that destination. You'll likely discover that you can take several routes in prompting to get you there. Sort and choose from your prompt strategies according to the requirements that mean the most to you. Perhaps you're looking for the most efficient

prompting path (one that uses fewer tokens) or the path that allows the AI chatbot to be more creative (meaning it can stray further in its interpretations of the prompt and the data).

If you want to learn more about this prompting technique and a variety of approaches — like the Inverted V and decision theory — read *Decision Intelligence For Dummies*.

If you don't care to dive that deep into the science of decision-making and keeping AI on track, suffice it to say that beginning at the end will give you the best start in accomplishing what you want in both advanced prompting and content engineering.



TIP Iterative prompting is a good fit for this approach once you have a well-defined end goal. Start with a broad prompt and then progressively narrow the focus in subsequent prompts based on the chatbot's responses until you reach the output you desire.

Managing Data for Targeted Impact on Outputs

The way you present information and the kind of data you provide can dramatically shape the responses you receive from your GenAI model. Managing and supplementing data in your prompts is one way to significantly influence the outputs. You can do this in several ways:

- **Include context within your prompts.** By providing background information or specifying the framework in the prompt, you can shape its responses better. For example, if you're asking about renewable energy, mentioning whether you're interested in its environmental impact, economic benefits, or technological challenges will steer a GenAI chatbot to consider data within those parameters and focus on the relevant aspects. Another way is to use clarifying follow-up questions in subsequent prompts. Consider asking for more details, requesting examples, or asking the chatbot to compare and contrast concepts.
- **Supplement data in your prompts** by providing specific examples, scenarios, reports, or data points for the GenAI model to consider. If you're discussing a complex issue like climate change, you might provide a hypothetical scenario or a case study, or name specific sources for the model to use.



REMEMBER These hints will point the *large language model* (LLM, the deep-learning base structure for GenAI) to the angle you're interested in exploring. If you prompt an LLM to provide data and resources in its responses, be sure to include a command for it to cite its sources. This will help you not only to fact-check responses but also to dive deeper into the details. Adding constraints to prompts by setting boundaries is another powerful technique. For example, you might ask for a summary suitable for a child, a detailed report for an expert, or a persuasive argument for a skeptic.

- **Incorporate recent data or findings** that may not be within the model's training data as a valuable strategy. Because GenAI models are trained up to a certain point in time, they may not have access to the latest information. By providing updated facts, figures, or developments in your prompt, or by using a chatbot that is connected to the internet, you can make responses as current and relevant as possible.
- **Use prompts to correct or counteract biases.** If you notice that the model's responses seem to lean in a certain direction, you can prompt the app to consider alternative perspectives or to provide a more balanced viewpoint. This is particularly important in fields in which new research or social changes might shift the understanding of a topic.



REMEMBER Think of your prompts as tools to sculpt an LLM's responses. Be mindful of the information you include, the questions you ask, and the constraints you set. This will help you to sharpen the model's focus and tailor its outputs to your specific needs and objectives. Remember, the more targeted your prompts, the more targeted the outputs will be.

You have two more sophisticated methods to manage and supplement data to shape outputs: providing supplemental data in custom instructions and using a technique known as *retrieval-augmented generation*, or *RAG*.

- **Incorporating supplemental data directly in your custom instructions can be a game-changer.** As mentioned earlier in this section, embedding additional information, such as recent statistics or findings, within your prompts is effectively updating the LLM's knowledge base.

However, this approach has limitations. The LLM can interpret and use only the data you provide within the scope of its abilities, and it may not always understand or apply the data correctly, especially if it's complex or requires expert analysis. Further, the data you provide in prompts is only applied to that thread or conversation. If you start a new chat, you need to enter that data in a prompt again for it to apply to that conversation, too.

One way to overcome this is to enter the data in custom instructions so it will be applied to all conversations until you change the instructions. In ChatGPT, you find custom instructions in the drop-down menu under your picture or icon at the upper-right corner of the user interface, as shown in Figure 6-1. In recent versions, you see the Customize ChatGPT option in the drop-down menu instead of the Custom Instructions option.

- ***Retrieval-augmented generation*, on the other hand, is like giving a GenAI chatbot a research assistant.** RAG allows the chatbot to pull in information from a large dataset or external sources in real time as it generates responses. This means that the model isn't just relying on the information it was trained on; it can access and use newer, supplemental data to inform its responses.

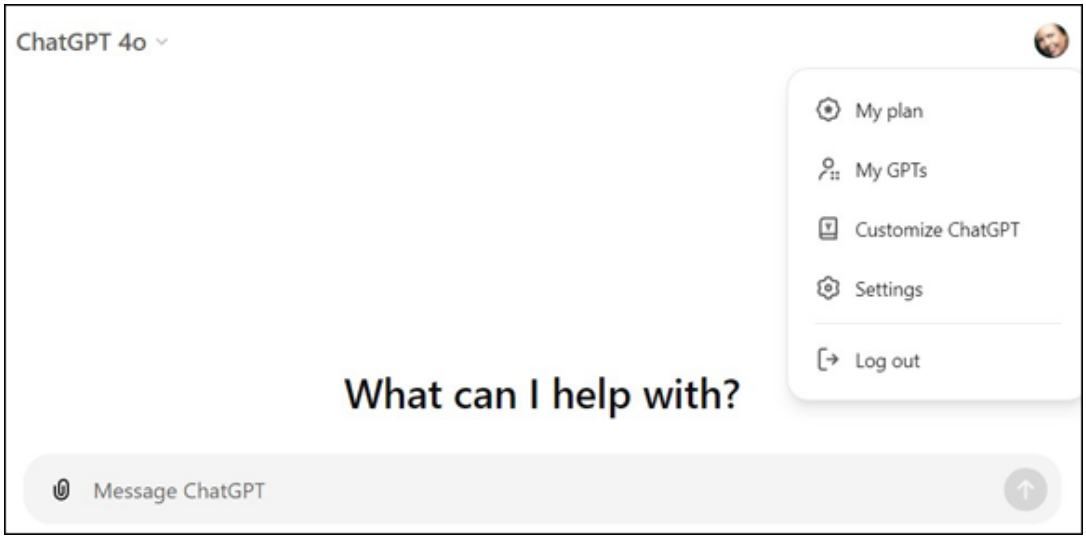


FIGURE 6-1: Customize ChatGPT from the drop-down menu in the upper-right corner of the window.

However, RAG also comes with limitations. The quality of the model's output depends heavily on the quality of the external sources it uses. If those sources are biased or inaccurate, the responses reflect that. Also, if the data is in nonstandard formats such as macro templates, spreadsheets, or proprietary formats, RAG in a GenAI chatbot might not directly accept or handle these forms. RAG works best with plain text or data that can be transformed into a textual format.

If neither of these methods meets your needs, you might consider looking for a specialized GenAI application designed for your specific domain, which would have been trained on a dataset that's more relevant to your

field or area of interest. This could yield better results without the need for extensive customization or supplementation.

Another option is to ask your IT department or technical team to refine the model itself. This could involve retraining the AI on a dataset that you provide, define, or choose. However, this is a resource-intensive solution that requires expertise in machine learning and AI, and it's not feasible for every situation.

Adding Data to Prompts

You've already discovered that one reason you might want to add data to your prompts is to guide an LLM toward a specific context or to ensure that the response takes into account the most current information, which might be outside of its original training data. But the placement of data within a prompt can also significantly affect the quality and relevance of the LLM's response.

You can think of the strategic placement of different kinds of data as setting the stage for a play. You need to place each piece of information in an order much like a character whose entrance and position on stage can influence the unfolding of the story.



TIP In short, best practice is to weave data throughout your prompt in a way that feels organic and logical, much like you're telling the chatbot a well-written story that includes your question or instructions at the end. Start by setting the stage, introduce key details where they have the most impact, and conclude with the direction you want the LLM to take.

However, it's important to stay mindful of GenAI's limitations. Overloading the prompt with too much data or excessively technical details can confuse the model and result in irrelevant, wrong, or incomplete responses. The art in effective prompting lies in providing just enough data to inform and guide the LLM's response without overshadowing your main question or request.

Using image inputs in ChatGPT prompts

Adding images to your prompt, if allowed by the AI chatbot you're using, is a good way to add data to your prompts without having to type all this information in the prompt bar. In ChatGPT, for example, you can click on the paper clip icon on the left side of the prompt bar to reveal the pull-down menu you see in Figure 6-2.

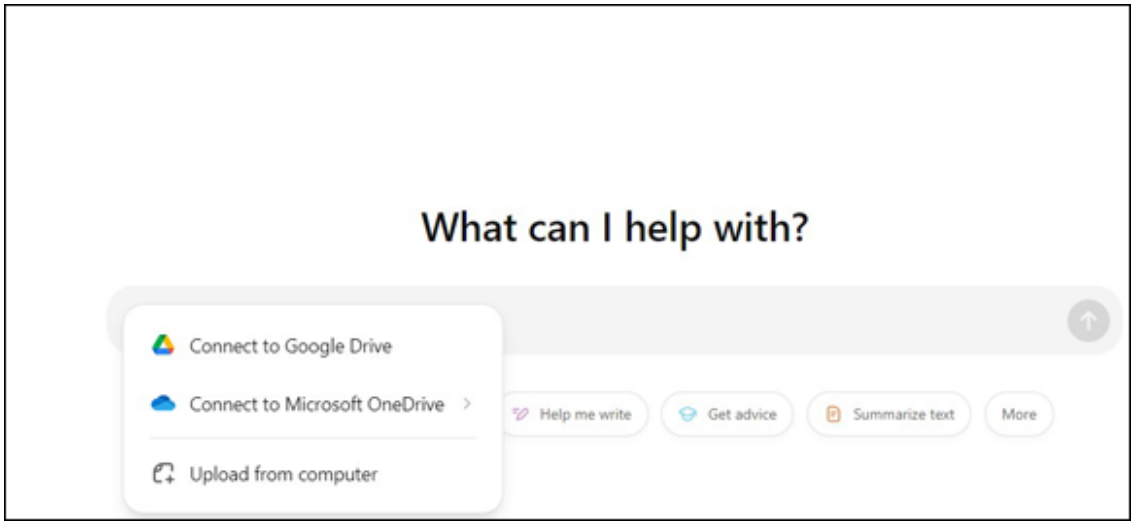


FIGURE 6-2: You can add files to the prompt bar by selecting the Upload from Computer option in the pull-down menu.

Most chatbots are limited in how much they can work with in regard to prompt attachments, however, so don't stuff too much in there. Generally, the same advice I gave you earlier on making prompts concise applies to the prompt attachments. In other words, don't upload a long document when you really want the model to use only a page or two.

Pictures as prompt input

But you can do much more in the way of prompting by using images and other attachments in a more strategic manner. For example, I can take a pic of a page from a book and prompt ChatGPT to develop a plan from that information. I've done so and depicted the steps in the figures to follow so you can see how this works.

First, I selected and photographed page 134 from *Generative AI For Dummies* about how to use AI aggregation to make stronger and more appealing content. You can see that in Figure 6-3.

Next, I attached it to the prompt bar by clicking on the paper clip icon on the left side of the bar. Then I added minimal instructions via text to the prompt bar. The part of the prompt that I typed reads:

> *Make a plan to design an online ad from the information in this image.*

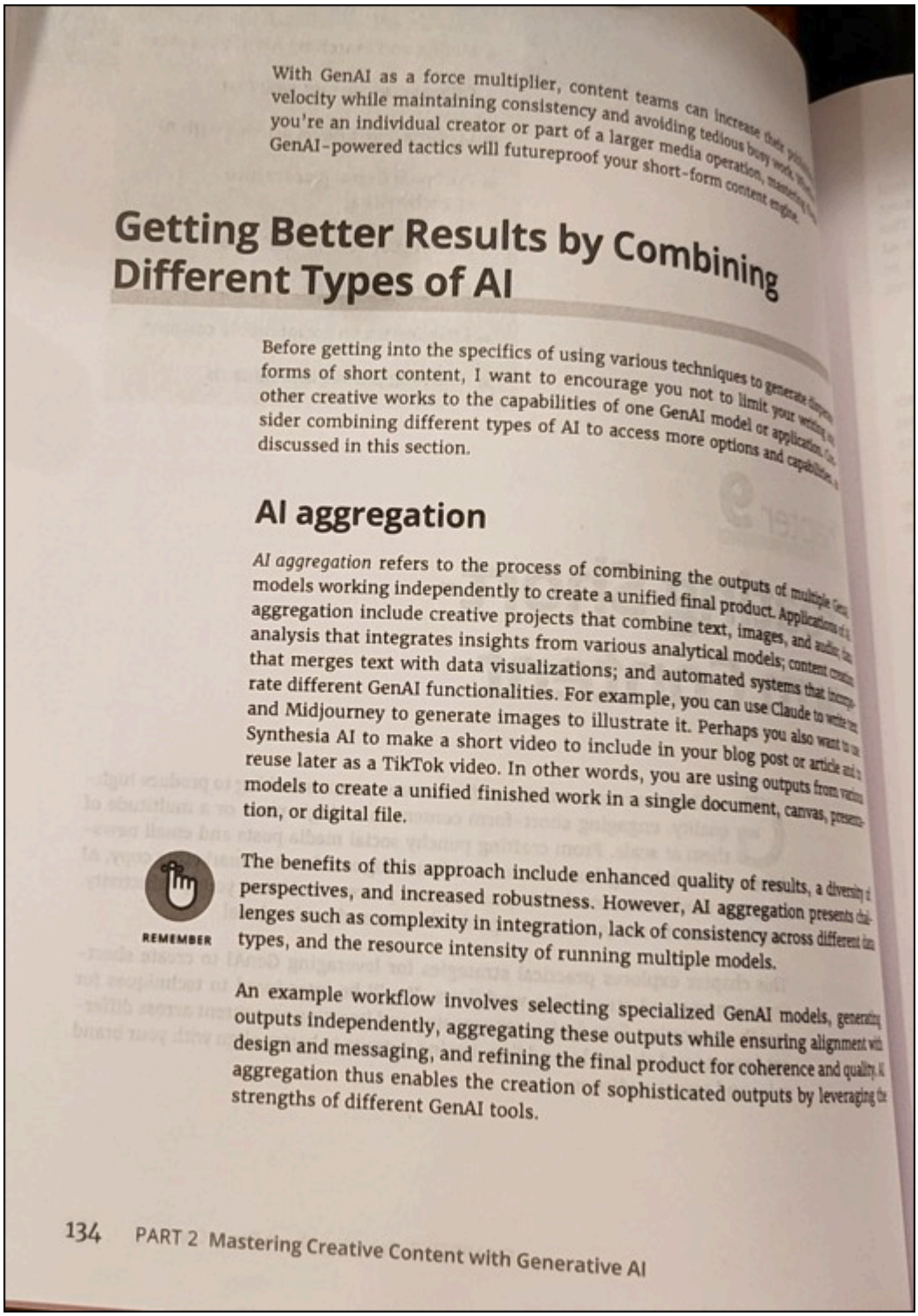
You can see that in Figure 6-4.

I then initiated the prompt by clicking on the arrow in the prompt bar. You can see the beginning of ChatGPT's response in Figure 6-5.

Random brilliant thoughts as prompt input

But that's just one example. Here's another. Imagine that you're attending a professional conference and are currently at a social event. Someone there makes an offhand comment that sparks a brilliant idea. You jot it down quickly on a napkin, as I've simulated in Figure 6-6.

You can take a picture of it and use it in a prompt to get ChatGPT to flesh out your idea, suggest additional or alternative ideas, point to factors involved that you need to consider, plan to implement your idea, and so on. Prompting with images and attachments truly is versatile. In Figure 6-7, you can see my full prompt for this demonstration.



With GenAI as a force multiplier, content teams can increase their
velocity while maintaining consistency and avoiding tedious busy work, whether
you're an individual creator or part of a larger media operation, mastering
GenAI-powered tactics will futureproof your short-form content engine.

Getting Better Results by Combining Different Types of AI

Before getting into the specifics of using various techniques to generate different
forms of short content, I want to encourage you not to limit your writing or
other creative works to the capabilities of one GenAI model or application. Con-
sider combining different types of AI to access more options and capabilities
discussed in this section.

AI aggregation

AI aggregation refers to the process of combining the outputs of multiple Gen
models working independently to create a unified final product. Applications of
aggregation include creative projects that combine text, images, and audio; data
analysis that integrates insights from various analytical models; content creation
that merges text with data visualizations; and automated systems that incorpo-
rate different GenAI functionalities. For example, you can use Claude to write text
and Midjourney to generate images to illustrate it. Perhaps you also want to use
Synthesia AI to make a short video to include in your blog post or article and to
reuse later as a TikTok video. In other words, you are using outputs from various
models to create a unified finished work in a single document, canvas, presenta-
tion, or digital file.

REMEMBER

The benefits of this approach include enhanced quality of results, a diversity of
perspectives, and increased robustness. However, AI aggregation presents chal-
lenges such as complexity in integration, lack of consistency across different data
types, and the resource intensity of running multiple models.

An example workflow involves selecting specialized GenAI models, generating
outputs independently, aggregating these outputs while ensuring alignment with
design and messaging, and refining the final product for coherence and quality. AI
aggregation thus enables the creation of sophisticated outputs by leveraging the
strengths of different GenAI tools.

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FIGURE 6-3: An image of page 134 of *Generative AI for Dummies*.

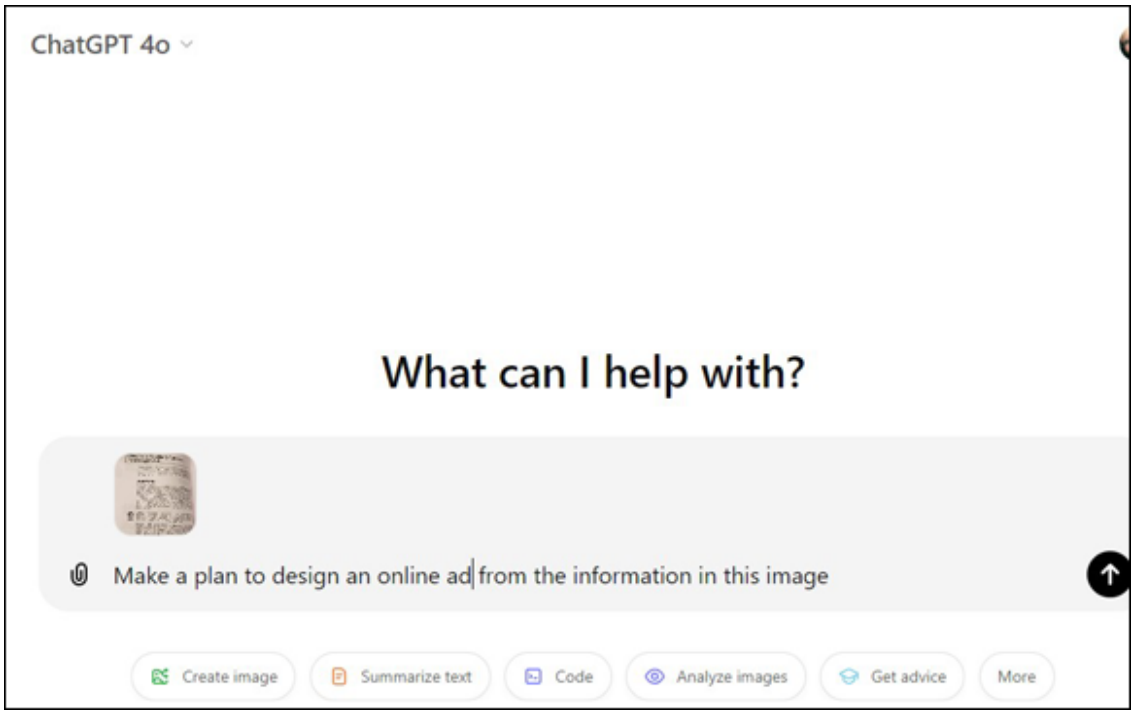


FIGURE 6-4: Adding the prompt after attaching an image.

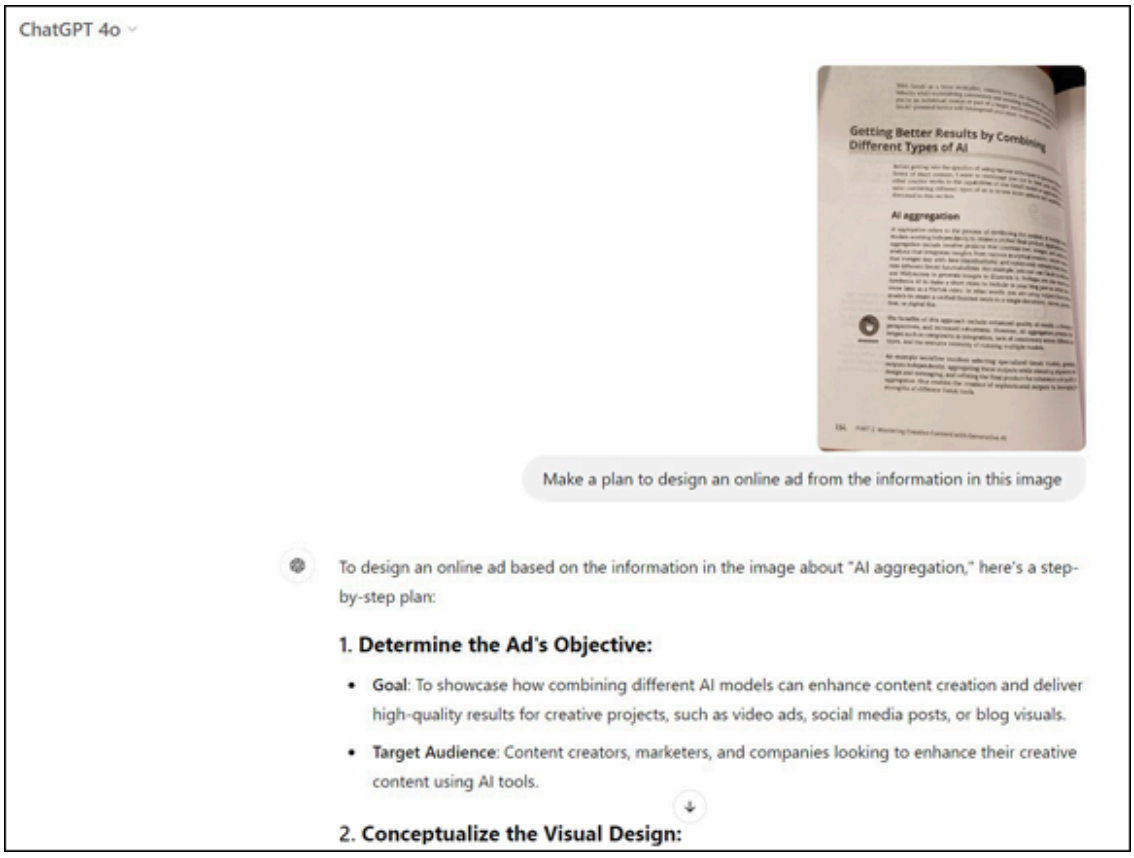


FIGURE 6-5: The output that ChatGPT generates after you click the arrow in the prompt bar.

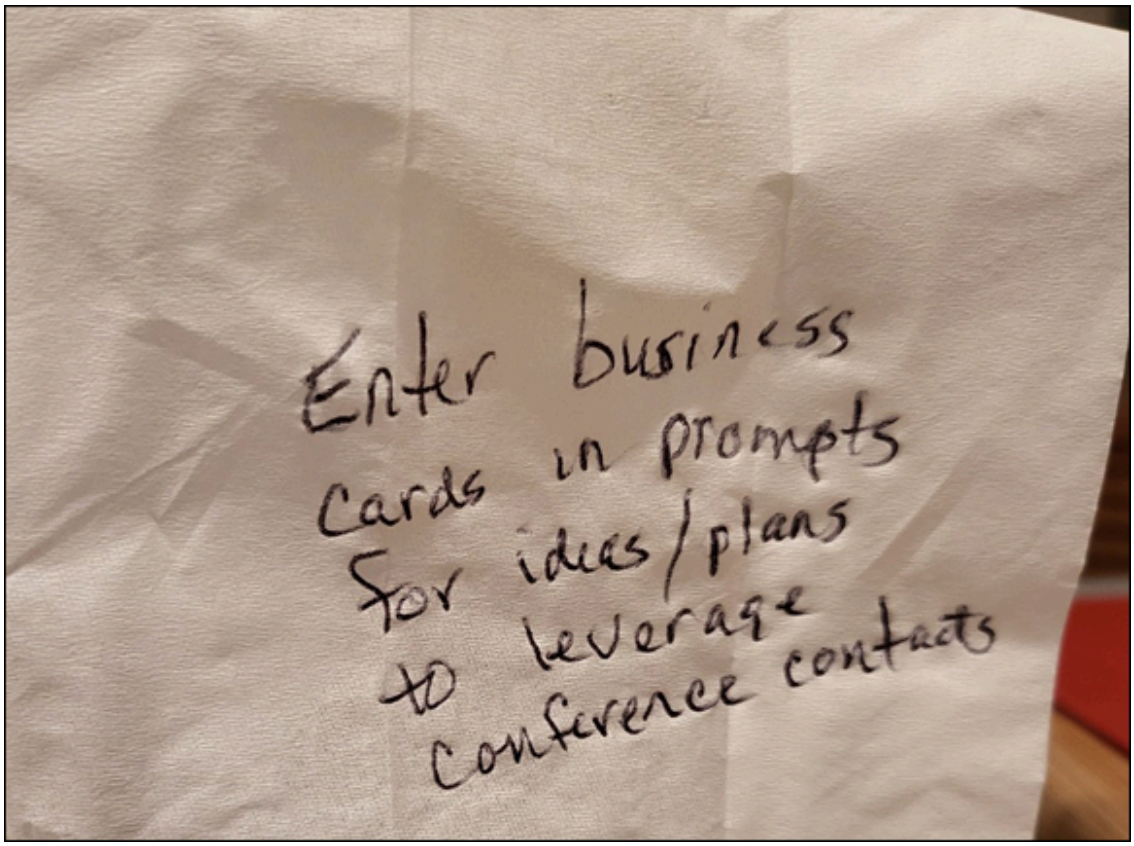


FIGURE 6-6: A handwritten note on a napkin can be later entered into ChatGPT with your prompt.

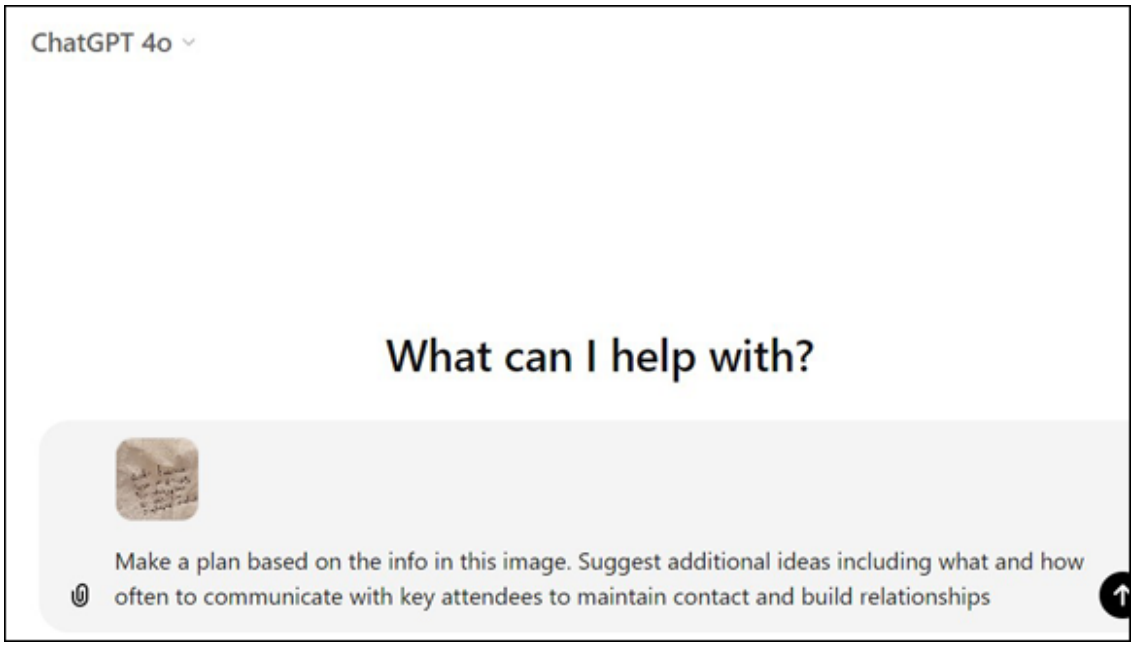


FIGURE 6-7: The prompt asks ChatGPT to plan based on the napkin image from Figure 6-6.

As before, I'm including a screenshot of ChatGPT beginning to respond to this prompt in Figure 6-8.

Refining with chained prompts

There's no reason for me to stop after the response to my last prompt. I can

- **Copy each of the tasks** in the list ChatGPT generates (its plan) and use them as prompts to get ChatGPT to do the work for me.

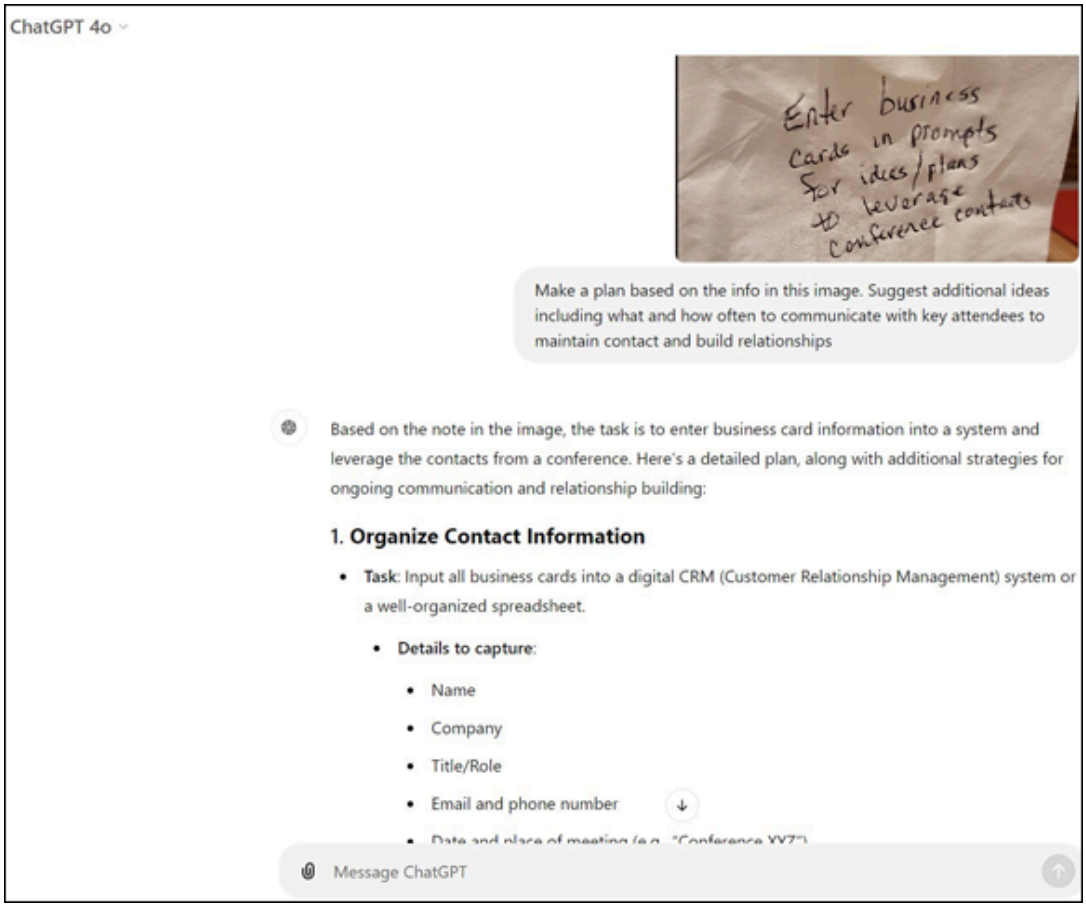


FIGURE 6-8: The response from ChatGPT after entering the prompt and uploading the image.

- **Use *chain prompting*** (a technique in which you use a series of prompts in sequence, each building upon the last to gradually refine or expand on the LLM's responses) to get ChatGPT to execute the tasks one at a time.
- **Take a screenshot or picture** of just one part of the plan that I actually want (scrapping the rest of ChatGPT's listed tasks), and on some future date, attach it to a prompt for ChatGPT to complete the work.



REMEMBER How you choose to do any of this refinement is up to you, and whatever guidelines or rules your employer wants you to follow. The point here is that ChatGPT's flexibility allows you to work in almost any way you want.

You can use this tactic for any project and any purpose. Consider using it to improve an image you've created in a generative AI tool, whether it's within ChatGPT or some other tool. For this demonstration, I created a simple black-and-white image in Craiyon, an AI-generated art platform, with a prompt instructing it to create an image that's wrong in several ways. The image I chose for this is shown in Figure 6-9.



FIGURE 6-9: One of nine images that Craiyon offered from my initial prompt.

Next, I attached the image in a ChatGPT prompt and instructed it to find and disclose the errors within the image and suggest ways to correct them. Figure 6-10 shows the actual prompt and part of ChatGPT's response.

These are only three simple examples of an endless number of possibilities in how adding image inputs to your prompts can work. My point here

is that prompting a GenAI model isn't limited to the words you type or to any preset formulations. Your ideas and imagination determine what you can do with GenAI.

Adding information to memory in ChatGPT

You can tell some chatbots to remember something, and they will! ChatGPT can remember any information that you prompt it to store in its memory (such as your preferences or details) across multiple conversations, not just within a single session. This allows ChatGPT to apply this information to existing and new chats, even if you come back later or after a long break.

When you tell ChatGPT to remember something in a prompt, it replies with a verbal confirmation and an icon saying, Memory Updated, as you see in Figure 6-11.

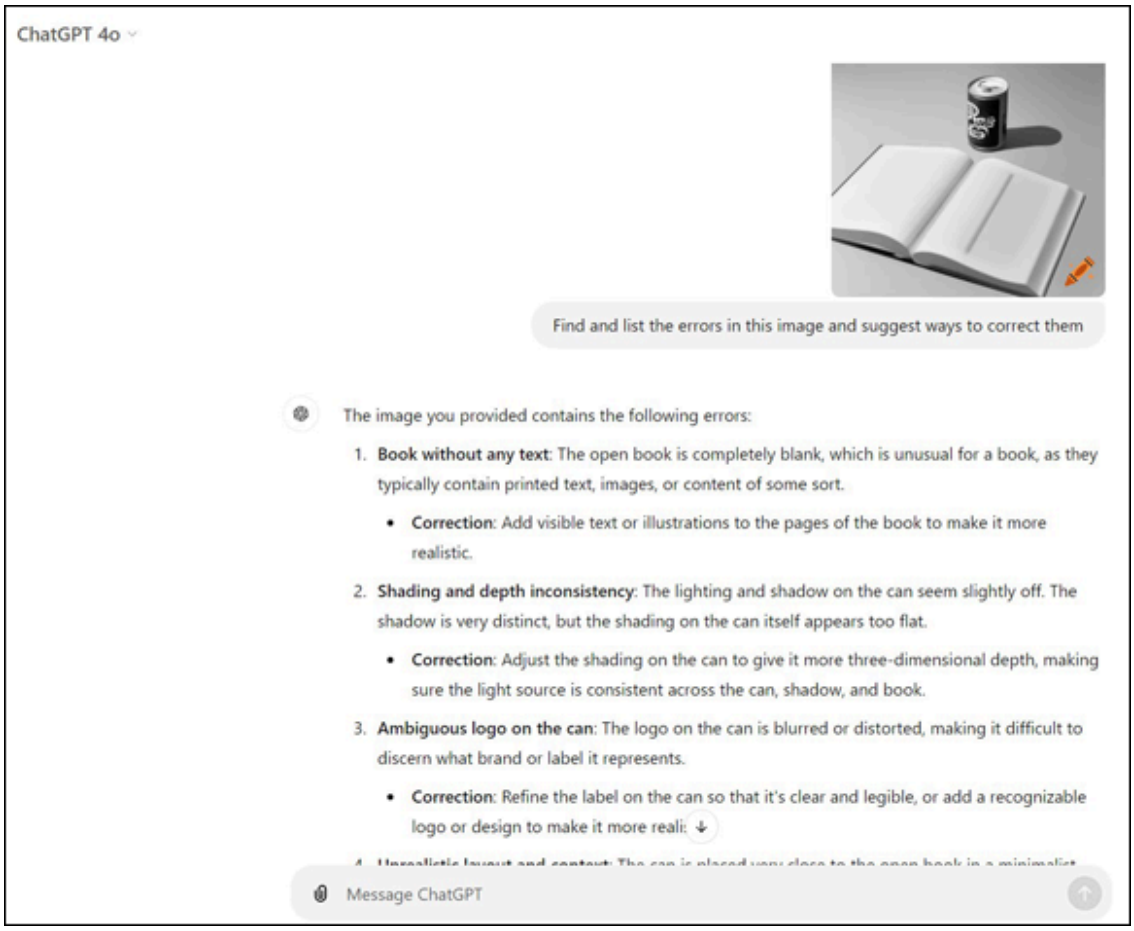


FIGURE 6-10: An uploaded image along with a prompt that asks ChatGPT to find errors in the image.



FIGURE 6-11: ChatGPT responds after being prompted to remember a name.

If you want to add something else for ChatGPT to remember, just tell it to do so in another prompt. Or if you want it to forget something you told it to remember earlier, tell it what to forget in a prompt.

This is the point at which you're probably wondering why you'd ever enter info into Custom Instructions or Customize ChatGPT fields if you can just tell ChatGPT to remember that info in a prompt. The answer is that using custom instructions and directly telling ChatGPT to remember something in a prompt serve slightly different purposes.



REMEMBER You set up custom instructions once, and those stay in constant use unless you change them manually. On the other hand, storing data in memory via a prompt allows you to dynamically make updates during conversations. For example, you can ask ChatGPT to remember or forget details as you go.

Here's why you might use custom instructions:

- **Long-term adjustments:** Custom instructions let you set up a general, ongoing framework for how you'd like ChatGPT to respond across all conversations. For example, if you have a preferred way for ChatGPT to explain things, custom instructions can ensure your wishes are consistently applied.
- **Consistency across sessions:** Custom instructions apply across multiple conversations even if you don't specifically mention something in a prompt. They shape the tone, detail, or context of every response, even if you forget to tell ChatGPT each time.

- **Specific guidelines:** If you want certain preferences to be remembered (like how much detail you want in answers or your usual areas of interest), custom instructions help keep things in check without your needing to remind the app each time.

In contrast, here's why you might use prompts to tell ChatGPT to remember something:

- **Specific to a conversation:** It's great for things that are important within a single chat or series of chats, and it allows you to be more spontaneous. It's also quick to undo when you're done since you can tell it to forget in a prompt, too.
- **Flexible and temporary:** If you just need ChatGPT to remember something for the short term, it's faster and easier than going into settings.

Both approaches work well, but it depends on how often you want to maintain certain preferences and how often you want to change things in memory on a whim.

Manipulating memory in ChatGPT

Now that you know how and where to add and delete information from ChatGPT's memory, it's time to look at how and why you might want to do so. You might recall that the key difference between memory and custom instructions is in the degree of flexibility. Memory is ideal for real-time quick adjustments in prompts, but custom instructions are better suited for establishing a consistent baseline. Although memory allows you to make changes on the go during chats, custom instructions stay the same until you manually update them in the settings.

Promoting consistent results

You can leverage these strengths of memory and custom instructions in several ways. For example, you might want to enter specific and more detailed information in custom settings to maintain consistency in results across chats on a large project.

For example, if you're writing a long academic paper, white paper, report, or book by prompting and writing in chunks rather than in one long prompt and response, it may be worth adding longer and more detailed

instructions in custom settings to serve as the guidelines for every chat and response until that project is finished. Typically, instructions added to memory in prompts are quick and short, so they don't carry as many details as custom instructions can.

However, you're limited on how much information you can store in custom instructions. The input fields for custom instructions are generally designed to accommodate a reasonable amount of text, but they're not intended for large amounts of detailed or complex information. Typically, each field for custom instructions — such as, "What would you like ChatGPT to know about you?" or "How would you like ChatGPT to respond?" — can hold a few paragraphs of text. This allows you to provide sufficient information for some customization, but it isn't suitable for storing large documents, extensive data, or complex lists. You'll want to put those items in retrieval-augmented generation (RAG) instead.

You're also limited to how much information you can store in memory. I can't tell you the exact capacity because that depends on the platform and the version you're using. But memory is designed to handle a reasonable amount of information to help personalize interactions; it's not intended for storing large or highly detailed datasets any more than custom instructions or customize ChatGPT features are.

Here are some general guidelines on memory limits:

- **Reasonable scope:** ChatGPT can remember personal preferences, ongoing projects, interests, and specific details you give it. But if you try to store too many details at once or extremely large chunks of text, ChatGPT won't remember it all.
- **Complexity:** Memory is best used for key pieces of information that tailor future conversations, such as your preferred learning style, favorite programming language, or recurring tasks. Highly complex or detailed information, like entire essays, large lists of technical specs, or multiple documents, may exceed the memory capacity. Again, it's better to add those items to RAG.
- **Updates over time:** You can add, update, or remove information in memory as needed, which allows you to keep it current without overloading it with too much at once.

Testing ChatGPT memory

If you aren't sure what ChatGPT is currently remembering for you, prompt it to give you a list.

If you're unsure about whether ChatGPT can retain certain details, you can test it. Just ask ChatGPT to remember those details, and it will let you know whether that's possible.



TIP Instead of thinking of this as an either/or option, strategically use information in custom instructions and memory so that you overcome the limits of both by making them work together to produce the results you seek.

Here are examples of manipulating information to get memory and custom instructions working in tandem:

- **Memory command in prompt:** "Remember that I'm working on a new project involving AI ethics."
- **Custom instructions update:** You could set custom instructions like, "I want ChatGPT to provide detailed explanations with examples whenever discussing complex topics" and "Cite sources complete with URLs in responses that include facts, numbers, or statistics."
- **Memory command in prompt:** "Remember that I'm currently working on a paper targeting an audience of pediatricians."
- **Custom instructions update:** "Write all responses using medical terminology and formal language suitable for an audience of physicians."



WARNING Speaking of memory, be cognizant of your chat history. GenAI chatbots automatically store your chats so you can return to them later as though no time has lapsed. However, the chatbot has no memory of previous chats. That's odd, I know, because you just learned about ChatGPT's memory. But ChatGPT remembers only what you specifically tell it to remember. It doesn't remember any of the chats stored in your chat history in the sidebar on the left side of the user interface.

Managing chat history

You might want to delete chats from your chat history for privacy and organization reasons. If you've shared sensitive or personal information about yourself, your customer, or your employer, removing the chat ensures that those details are no longer stored. It doesn't, however, necessarily prevent that data from being harvested to train other AI models or possibly leaking into responses the LLM delivers to other users. It's best if you never enter sensitive information into chats.



REMEMBER As your chat history grows, it can become cluttered with old or irrelevant conversations. Deleting these older chats can help keep things organized, making it easier to navigate and find the conversations that matter most to you.

Consider these other reasons for deleting or otherwise managing your chat history:

- **The desire for a fresh start.** If you've been working on a particular project or topic and want to shift your focus, removing past chats can help you feel like you're starting anew. Similarly, if previous conversations contain outdated or incorrect information, clearing them out can prevent confusion going forward, allowing you to reset your chat history with current, accurate information.
- **Avoiding potential misunderstandings or misuse of certain conversations.** You might want to delete chats that could be misinterpreted, no longer reflect your views, or can be used against you in the future. Remember that if you're using an enterprise version of an AI chatbot, your employer can likely see all your prompts and responses. To prevent future complications, don't say anything in a prompt that you wouldn't say to your boss, and if you have done so already, you might want to delete that now.
- **Eliminating more personal discussions, even if they aren't sensitive.** Deleting them can offer peace of mind and make you feel more in control of your environment.
- **Archiving chats or deleting your entire chat history at once.** You find the necessary command buttons under settings on the user interface (found under your profile picture in ChatGPT). In some versions, you also find a "clear all" button at the bottom of your chat history.

Changing the Model's Temperature

Changing the model's temperature means you're adjusting an LLM's creativity or randomness level. A *lower temperature setting* results in more predictable, conservative, and determinate outputs. This means the model will stick close to the most likely responses based on its training. It's useful when you want information that's straightforward, factual, and meets general expectations, including yours, your boss's, and your audience's.

On the other hand, a *higher temperature setting* allows the model to take more risks in its responses. Make it hot in here, and the model will take liberties left, right, and center in interpreting the data, which often leads to more creative or unusual outputs. You might want to cut an LLM loose like this when you're seeking out-of-the-box ideas, brainstorming, searching for a range of different perspectives, or trying to get its response closer to your own creative vision.

For example, if you're asking for a recipe for chicken soup, a lower temperature might give you a classic recipe, while a higher temperature might suggest a fusion of cuisines for an interesting spin on the traditional fare. Similarly, if you're asking for story ideas, a low temperature might give you familiar tropes, whereas a high temperature could provide you with a plot involving less conventional elements and unexpected endings.

To change the model's temperature, you, someone in your company's IT department, or an admin typically adjusts a parameter in the system that's controlling the model. This parameter usually ranges from 0 to 2, with 0 being the lowest temperature (most deterministic) and 2 being the highest (most random). The more balanced range are temps 0.5 to 1.0. You adjust this setting before making a request to the model. If you're using a customer-facing or consumer version of an LLM, you're not going to have a way to change this setting. But certain applications and platforms will let users change the model's temperature themselves.



TIP If you're unsure about what temperature to use, start with a moderate setting, like 0.5 or 1.0, and see if the responses meet your needs.

You can then adjust the setting based on whether you want more conventional or more creative responses.

Changing the Model's Weights

Changing the model's *weights* (the amount of influence that each input feature has on the output) involves updating or retraining a model's parameters. It's a technical method of affecting a model's outputs that most people won't have access or the skills to do. However, it's important for you to be aware of its existence if for no other reason than you know what to ask it — and how to articulate it — if you're having trouble with a chatbot's performance on the job.

You change a model's weights to enhance performance, adapt to new data, or correct biases. You might want to do this if the model is making less than accurate predictions, particularly if it's overfitting, underfitting, or not generalizing well to new data. *Overfitting* happens when a model learns the training data too well, capturing not just the underlying patterns but also the noise and minor details that are irrelevant. *Underfitting* is just the opposite and occurs when a model is too simple to capture any of the underlying patterns in the data.

When *data drift* (data changes over time that result in the AI model becoming ineffective) or *concept drift* (patterns change, which cause the model to become increasingly inaccurate over time) occurs, results can be wonky or wrong. That issue is common in areas such as finance or personalized recommendations where data changes quickly. Adjusting the weights allows the model to better reflect these new patterns. Similarly, if you notice the model behaving in biased ways, changing the weights through additional training or fine-tuning can minimize the issue.

You can change a model's weights by training a new model or retraining an existing one. Neither is particularly cheap or easy to do. Fine-tuning is another method of slightly adjusting the weights of an existing pretrained model, making it more suitable for a specific task without having to start a new model from scratch. Unfortunately, fine-tuning is another difficult and complex task.

Another way to adjust the model's learning behavior is by modifying the learning rate, which controls how much the weights change during training. If the model isn't learning well, adjusting the learning rate can ensure more effective updates to the weights.